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# Certainty Correspondence: Higher-Order Perturbation in the Sequence Space

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(based on work in progress with  
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Goethe Heterogeneous-Agent Macro Workshop, 2026



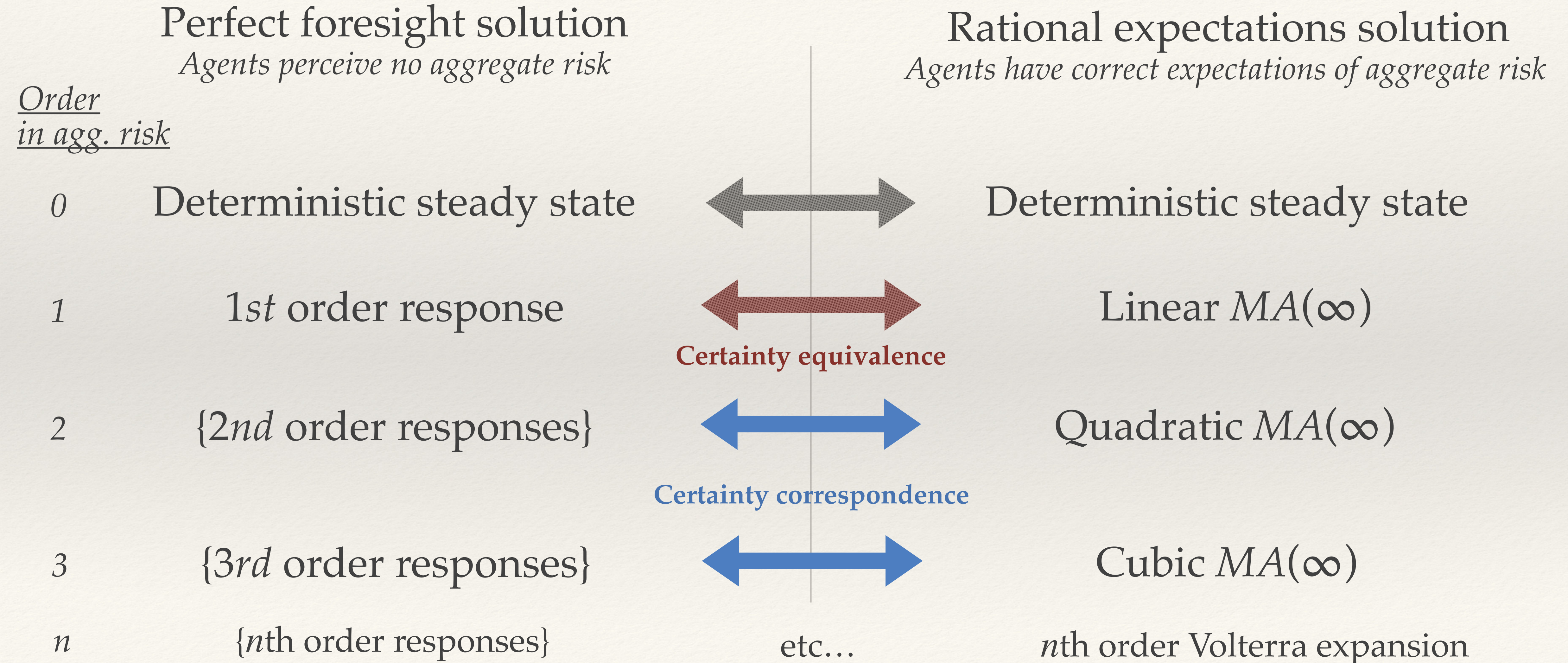
# Aggregate risk in heterogeneous-agent models

- ❖ Heterogeneous-agent (HA) models with aggregate risk are difficult to solve
- ❖ **Common strategy:** solve with perfect foresight, appeal to **certainty equivalence**
  - ❖ 1st order perfect foresight  $\Leftrightarrow$  1st order perturbation solution  
[e.g. Boppart-Krusell-Mitman, Auclert-Bardoczy-Rognlie-Straub,...]
- ❖ Critique: *“not suitable for Qs in which aggregate risk and non-linearities play a key role”*  
[Moll, 2024]
- ❖ This paper: **certainty correspondence !**
  - ❖ Perfect foresight still enough to get **2nd, 3rd (+) order perturbation solution!**
  - ❖ Obtain same answer as “pruned” state-space perturbation, but much faster!

[state-space perturbation as in Bhandari-Bourany-Evans-Golosov, Bilal, Bayer-Luetticke, Reiter...;  $\neq$  “global solution”]



# Certainty correspondence





Setup



# Setup (same as in lecture 8)

- ❖ Consider macro model:

$$F\left(X_{t-1}, X_t, \mathbb{E}_t[X_{t+1}], \epsilon_t\right) = 0 \quad \text{Contrast with usual formulation} \quad \mathbb{E}_t\left[F\left(X_{t-1}, X_t, X_{t+1}, \epsilon_t\right)\right] = 0$$

- ❖  $X_t \in \mathbb{R}^n$  are endogenous variables [het-agent applications:  $n$  large, eg  $n = 2,000$ ]
- ❖  $\epsilon_t \in \mathbb{R}$  are exogenous innovations (realized shock process) [paper extends to  $>1$  shock]
  - ❖ Distributed  $\epsilon_t = \sigma_R \cdot \bar{\epsilon}_t$  with  $\bar{\epsilon}_t$  iid, mean 0, variance 1, and symmetric density
- ❖ Expectations  $\mathbb{E}_t[\cdot]$  are taken assuming future distribution of  $\epsilon_t$  is  $\sigma \cdot \bar{\epsilon}_t$ 
  - ❖ **Rational expectations:**  $\sigma = \sigma_R$  (perceived distribution of future  $\epsilon_t = \text{truth}$ )
  - ❖ **Perfect foresight:**  $\sigma = 0$  (agents think all future  $\epsilon_t$  will realize to 0 for sure)
- ❖ **Deterministic steady state**  $X \in \mathbb{R}^n$  solves  $F(X, X, X, 0) = 0$  (i.e.  $\sigma = \sigma_R = 0$ )



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# (Stochastic) sequence space solution

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- ❖ Given  $\sigma$ , consider **nonlinear sequence-space solution** to  $F$ :

$$X_t = \mathcal{X} \left( \sigma; \epsilon_t, \epsilon_{t-1}, \epsilon_{t-2}, \dots \right) \quad [\text{Lan-Meyer-Gohde, Lombardo-Uhlig}]$$

[eg, obtained from stable state-space solution  $X_t = g \left( X_{t-1}, \epsilon_t; \sigma \right)$ ]

- ❖ Our paper relates this to the perfect-foresight solution

$$X_t^{PF} = \mathcal{X} \left( 0; \epsilon_t, \epsilon_{t-1}, \epsilon_{t-2}, \dots \right)$$

[eg, obtained from  $\infty$ -long sequence of repeated-surprise “MIT” shocks]

- ❖ Use **perturbation approach**: linearize  $\mathcal{X}$  around  $\sigma = \sigma_R = 0$  (i.e.  $\epsilon = \mathbf{0}$ )

- ❖ Helpful result:  $\mathcal{X} \left( \sigma; \epsilon_t, \epsilon_{t-1}, \dots \right) = \mathcal{X} \left( -\sigma; \epsilon_t, \epsilon_{t-1}, \dots \right)$ , so, e.g.,  $\mathcal{X}_\sigma(\mathbf{0}) = 0$



# MIT shock impulse responses

- ❖ Special case of  $X^{PF}$  is the MIT shock impulse response

$$X_j^{MIT}(\nu) \equiv \mathcal{X}(0; 0, \dots, 0, \underset{\substack{\uparrow \\ \text{position } j}}{\nu}, 0 \dots)$$

- ❖ Can compute  $\{X_j^{MIT}(\nu)\}_j$  with nonlinear perfect foresight starting from  $X$
- ❖ Similarly, for  $l \geq 1$  we have the sequence of two MIT impulse responses

$$X_j^{MIT,l}(\nu, \epsilon) \equiv \mathcal{X}(0; 0, \dots, 0, \underset{\substack{\uparrow \\ \text{position } j}}{\nu}, 0, \dots, 0, \underset{\substack{\uparrow \\ \text{position } j+l}}{\epsilon}, 0 \dots)$$

- ❖ For each  $l$ , can compute  $\{X_j^{MIT,l}(\nu, \epsilon)\}_j$  starting from  $X_l^{MIT}(\epsilon)$
- ❖ **Next:** why these are sufficient to get the second-order solution!



# Warm-up: certainty equivalence

- ❖ Expand  $X_t = \mathcal{X}(\sigma, \epsilon_t, \epsilon_{t-1}, \dots)$  to first order in  $(\sigma, \epsilon)$ :  
No anticipated risk, no past realization

$$X_t = X + \cancel{\mathcal{X}_\sigma \sigma} + \frac{\partial \mathcal{X}}{\partial \epsilon_0}(\mathbf{0}) \epsilon_t + \frac{\partial \mathcal{X}}{\partial \epsilon_{-1}}(\mathbf{0}) \epsilon_{t-1} + \dots + \mathcal{O}(\sigma^2)$$

Symmetry:  
Steady state unchanged

- ❖ So, to first order,  $X_t$  follows **linear MA** process, with coefficients equal to

$$\frac{\partial \mathcal{X}}{\partial \epsilon_{-j}}(\mathbf{0}) = \frac{dX_j^{MIT}}{d\nu}(0) \equiv \mathcal{X}_j$$

**Certainty equivalence.** To first order,  $X_t$  follows a linear MA process whose coefficients are given by the first order MIT shock impulse response of  $X$  to  $\epsilon$  [Boppart-Krusell-Mitman]



# Second-Order Certainty Correspondence



# Second-order sequence space perturbation

- ❖ Now expand to second order in  $(\sigma, \epsilon)$  around  $(0, \mathbf{0})$  (“Volterra expansion”)

- ❖ Use symmetry result again:  $\frac{\partial^2 \mathcal{X}}{\partial \sigma \partial \epsilon_{-j}}(\mathbf{0}) = 0$

- ❖ Find:

$$X_t = X + \sum_{j=0}^{\infty} \mathcal{X}_j \epsilon_{t-j} + \frac{1}{2} \left( \mathcal{X}_{\sigma\sigma}(\mathbf{0}) \sigma^2 + \sum_{j,k \geq 0} \frac{\partial^2 \mathcal{X}}{\partial \epsilon_{-j} \partial \epsilon_{-k}}(\mathbf{0}) \epsilon_{t-j} \epsilon_{t-k} \right) + \mathcal{O}(\sigma^3)$$

Still no anticipated risk  
or past realization

quadratic MA process

- ❖ Rearrange and interpret...



# Interpreting the quadratic MA process

Anticipated risk

Size dependence

State dependence

$$X_t \simeq X + \sum_{j=0}^{\infty} \mathcal{X}_j \epsilon_{t-j} + \boxed{\frac{1}{2} \mathcal{X}_{\sigma\sigma} \sigma^2} + \boxed{\frac{1}{2} \sum_{j=0}^{\infty} \frac{\partial^2 \mathcal{X}}{\partial \epsilon_{-j}^2} \epsilon_{t-j}^2} + \boxed{\sum_{j=0}^{\infty} \sum_{l=1}^{\infty} \frac{\partial^2 \mathcal{X}}{\partial \epsilon_{-j} \partial \epsilon_{-(j+l)}} \epsilon_{t-j} \epsilon_{t-(j+l)}}$$

- ❖ Size and state dependence are second-order MIT shock impulses

$$\frac{\partial^2 \mathcal{X}}{\partial \epsilon_{-j}^2} = \frac{d^2 X_j^{MIT}}{d\nu^2}(0) \equiv \mathcal{X}_{jj} \quad \frac{\partial^2 \mathcal{X}}{\partial \epsilon_{-j} \partial \epsilon_{-(j+l)}} = \frac{\partial^2 X_j^{MIT,l}}{\partial \epsilon \partial \nu}(0,0) \equiv \mathcal{X}_{jj+l}$$

- ❖ **Next:** How to solve for  $\mathcal{X}_{\sigma\sigma}$  using only perfect foresight as well!



# Anticipated risk term

- ❖ How do we compute **anticipated risk term**  $\frac{1}{2}\mathcal{X}_{\sigma\sigma}\sigma^2$  ?
- ❖ Idea: solve for the **risky steady state** with  $\sigma_R = 0$  but  $\sigma > 0$
- ❖ If all shocks have realized to  $\epsilon_{t-j} = 0$  in the past then we have:

$$X_t = X_{t-1} \simeq X + \frac{1}{2}\mathcal{X}_{\sigma\sigma}\sigma^2$$

**Risky steady state (unknown)**

$$\mathbb{E}_t X_{t+1} \simeq X + \frac{1}{2}\mathcal{X}_{\sigma\sigma}\sigma^2 + \frac{1}{2}\mathcal{X}_{00}\sigma^2$$

**Anticipation of impact size dependence (known)**

- ❖ Find this  $X^*$  by using steady state code to solve  $F\left(X^*, X^*, X^* + \frac{1}{2}\mathcal{X}_{00}\sigma^2, 0\right) = 0$

i.e. add impact size dependence  $\mathcal{X}_{00}\sigma^2/2$  to  $\mathbf{v}$  in equations where  $\mathbb{E}_t[\mathbf{v}_{t+1}]$  appears



# Bottom line: solving the quadratic MA process

Anticipated risk

Size dependence

State dependence

$$X_t \simeq X + \sum_{j=0}^{\infty} \mathcal{X}_j \epsilon_{t-j} + \boxed{\frac{1}{2} \mathcal{X}_{\sigma\sigma} \sigma^2} + \boxed{\frac{1}{2} \sum_{j=0}^{\infty} \mathcal{X}_{jj} \epsilon_{t-j}^2} + \boxed{\sum_{j=0}^{\infty} \sum_{l=1}^{\infty} \mathcal{X}_{j,j+l} \epsilon_{t-j} \epsilon_{t-(j+l)}}$$

Compute by modifying  
steady state equations

Compute with second-  
order MIT shock

Compute this with two  
successive MIT shocks,  $\times L$

❖ **Next:** Implications for model impulse responses and moments



# Generalized impulse response

- ❖ Define generalized IRF of  $X$  as

$$IRF_j \left( \nu, \left\{ \epsilon_{-k} \right\}_{k=1}^{\infty} \right) \equiv \mathbb{E} \left[ X_j \mid \epsilon_0 = \nu, \left\{ \epsilon_{-k} \right\}_{k=1}^{\infty} \right] - \mathbb{E} \left[ X_j \mid \left\{ \epsilon_{-k} \right\}_{k=1}^{\infty} \right] \quad j \geq 0$$

- ❖ We have:

First-order IRF

Size dependence

State dependence

$$IRF_j \left( \epsilon_0 = \nu, \left\{ \epsilon_{-k} \right\}_{k=1}^{\infty} \right) = \boxed{x_j \nu} + \boxed{\frac{1}{2} \mathcal{X}_{jj} (\nu^2 - \sigma^2)} + \boxed{\sum_{l=1}^{\infty} \mathcal{X}_{j,j+l} \nu \epsilon_{-l}}$$

Linear, not state  
dependent

Subtracts *expected*  
*size dependence*

Interaction of date-0 shock  
with shock  $l$  periods ago



# Model moments

- ❖ Ergodic (long-run) mean under rational expectations ( $\sigma_R = \sigma$ )

$$\mathbb{E}[X] = \underbrace{X + \frac{1}{2} \mathcal{X}_{\sigma\sigma} \sigma^2}_{\text{Risky steady state}} + \underbrace{\frac{1}{2} \sum_{j=0}^{\infty} \mathcal{X}_{jj} \sigma^2}_{\text{Expected size dependence}}$$

- ❖ Ergodic variance from 2nd-order *MA*:

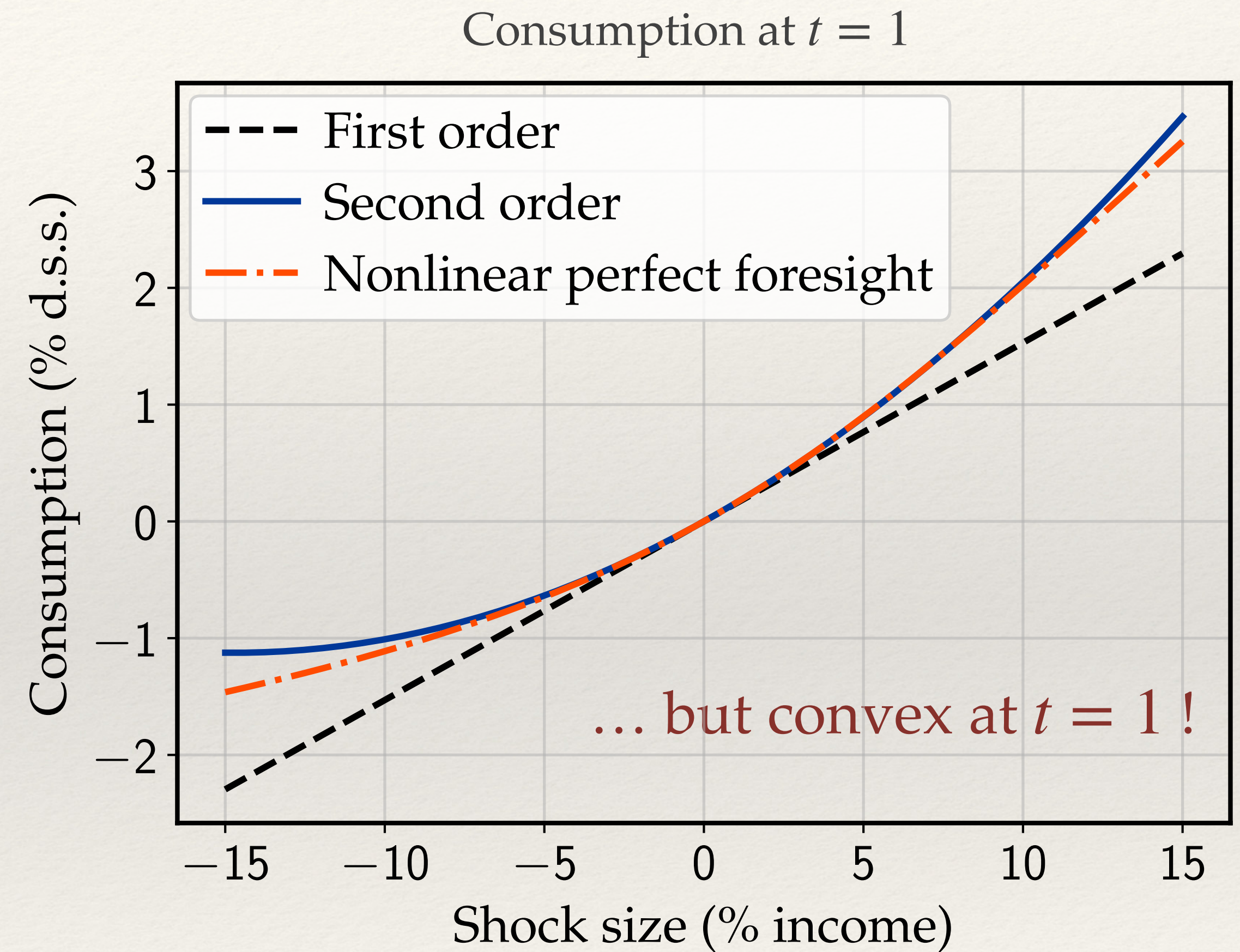
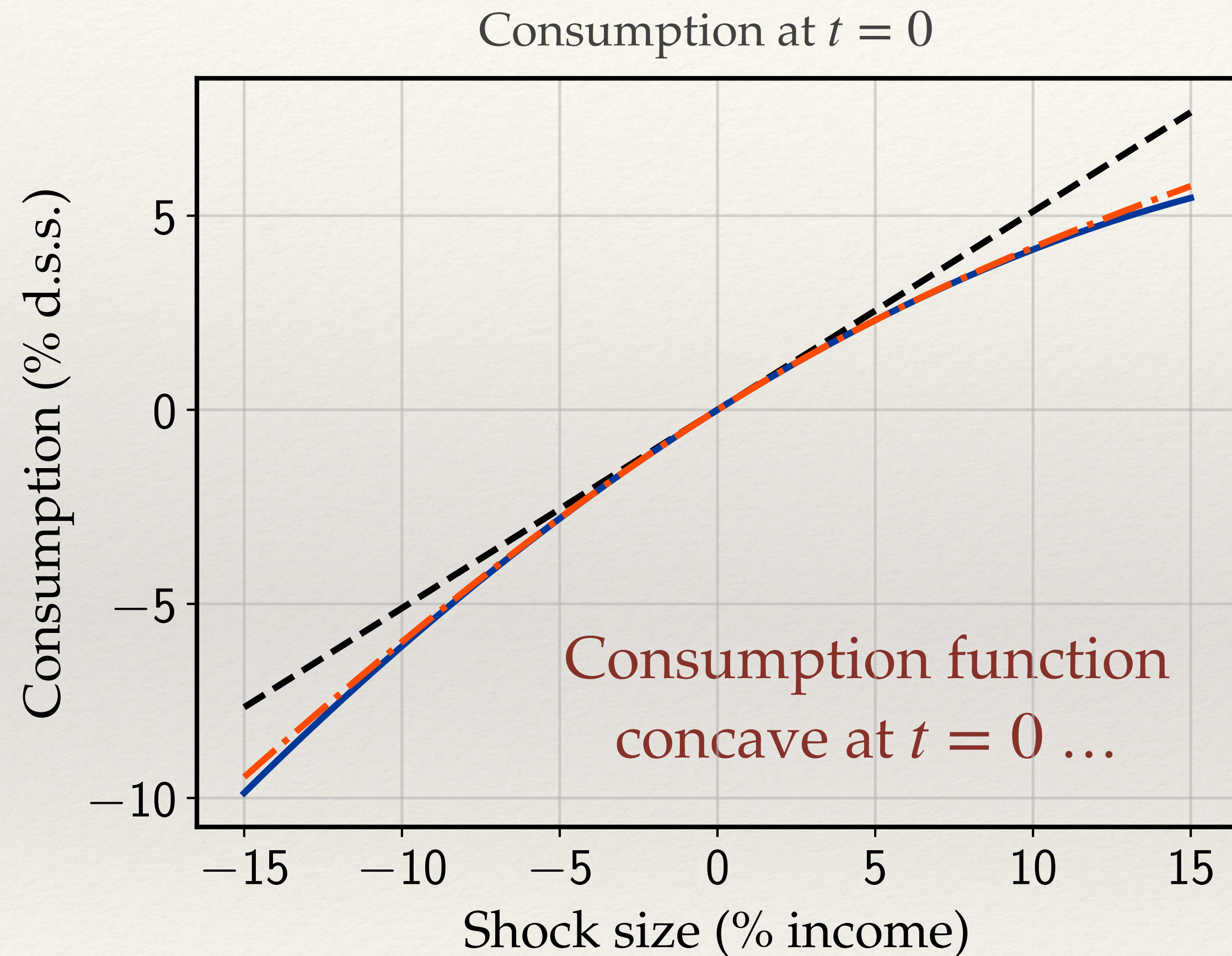
$$\text{Var}(X) = \underbrace{\sigma^2 \sum_{j=0}^{\infty} \left( \mathcal{X}_j \right)^2}_{\text{Implied by 1st-order MA (3rd order accurate)}} + \underbrace{\frac{\sigma^4}{2} \sum_{j,k=0}^{\infty} \left( \mathcal{X}_{jk} \right)^2}_{\text{Same as in pruned 2nd-order state-space}}$$

Implied by 1st-order MA (3rd order accurate)    Same as in pruned 2nd-order state-space

- ❖ In paper: 4th-order accurate  $\text{Var} \left( \mathcal{X}(\sigma; \sigma \bar{\epsilon}_t, \sigma \epsilon_{t-1}^-, \dots) \right)$  adds terms from 3rd-o. *MA*



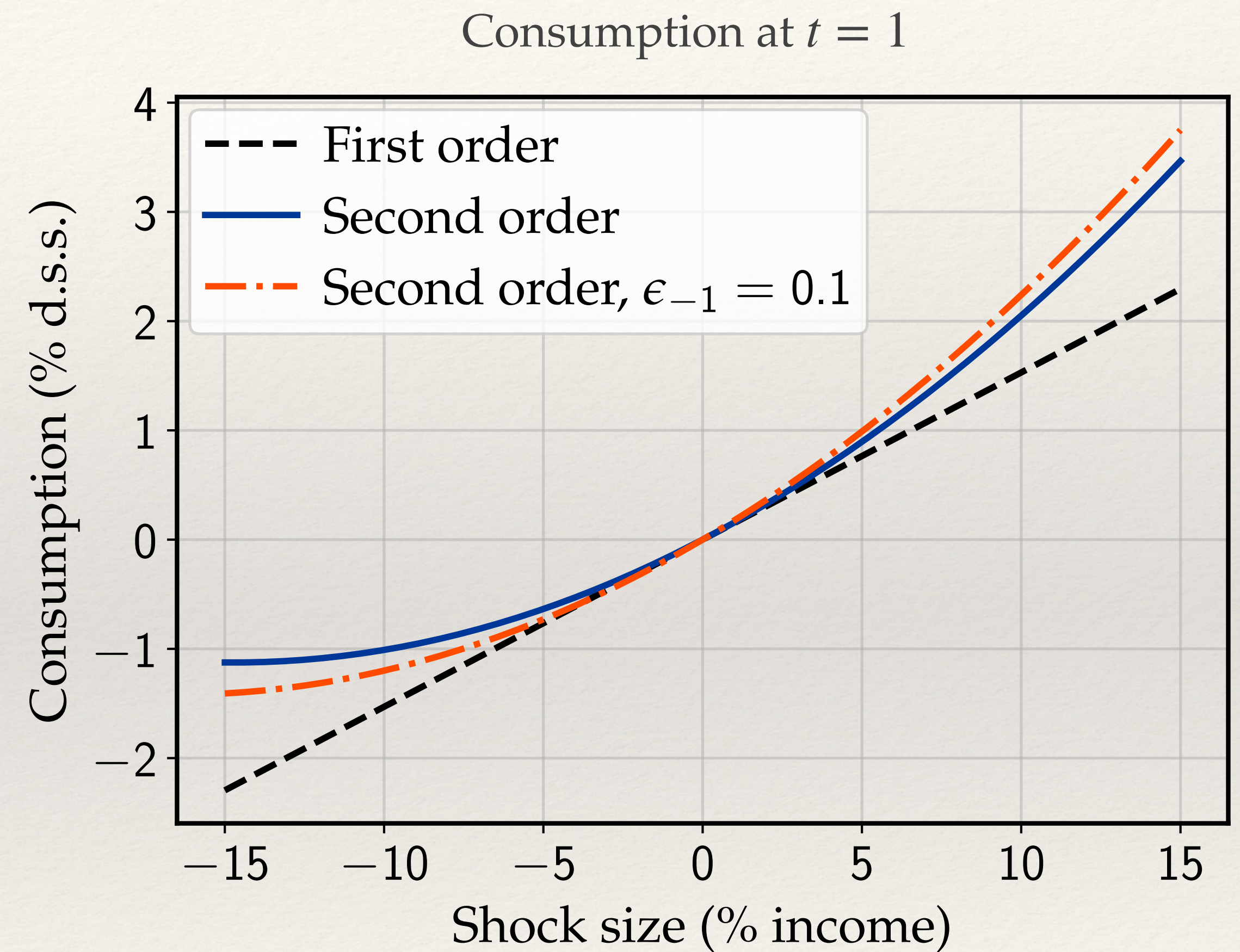
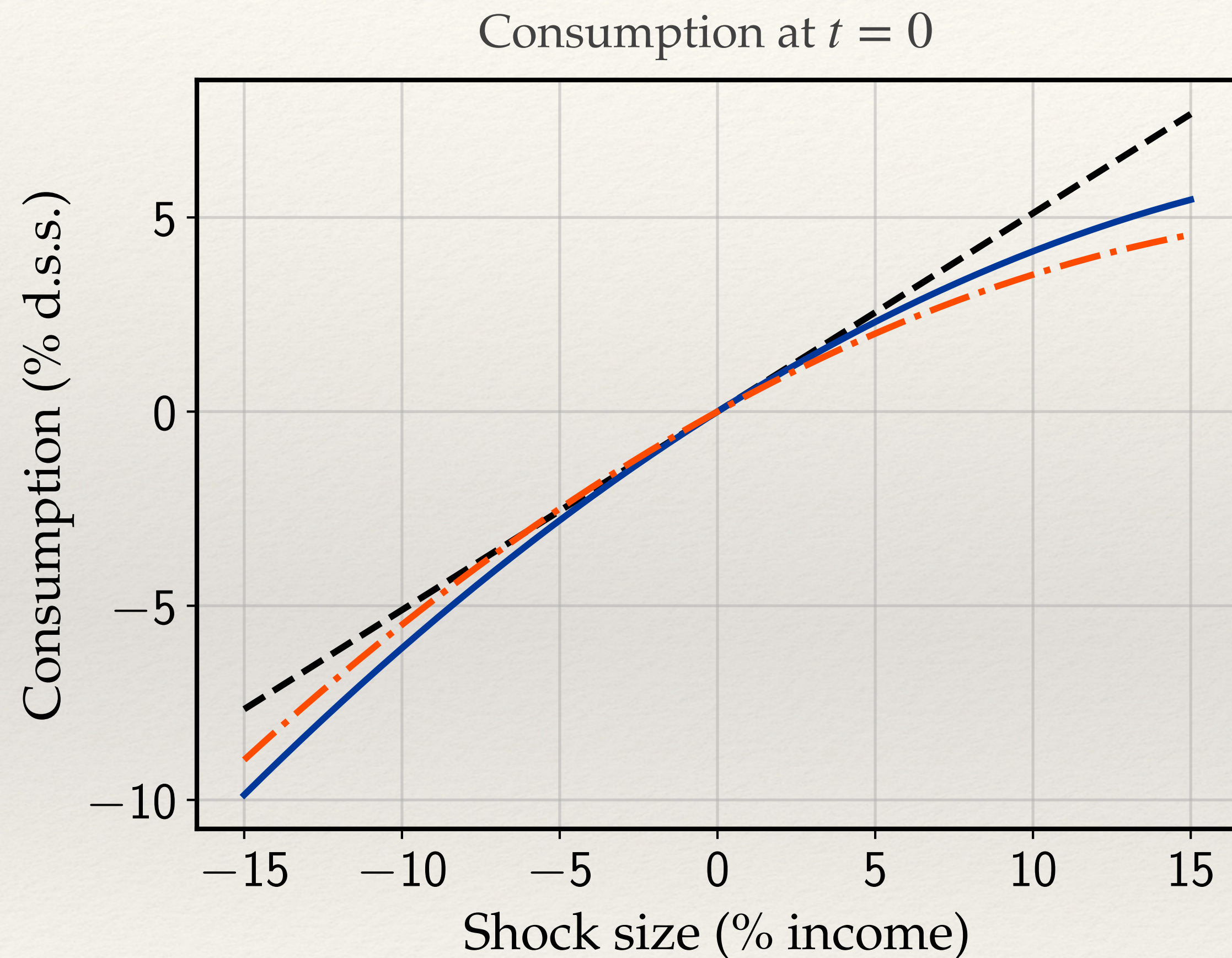
# Application: size dependence in consumption



❖ Nonlinear iMPCs! Large shock  $\rightarrow$  lower MPC today ... but higher tomorrow!



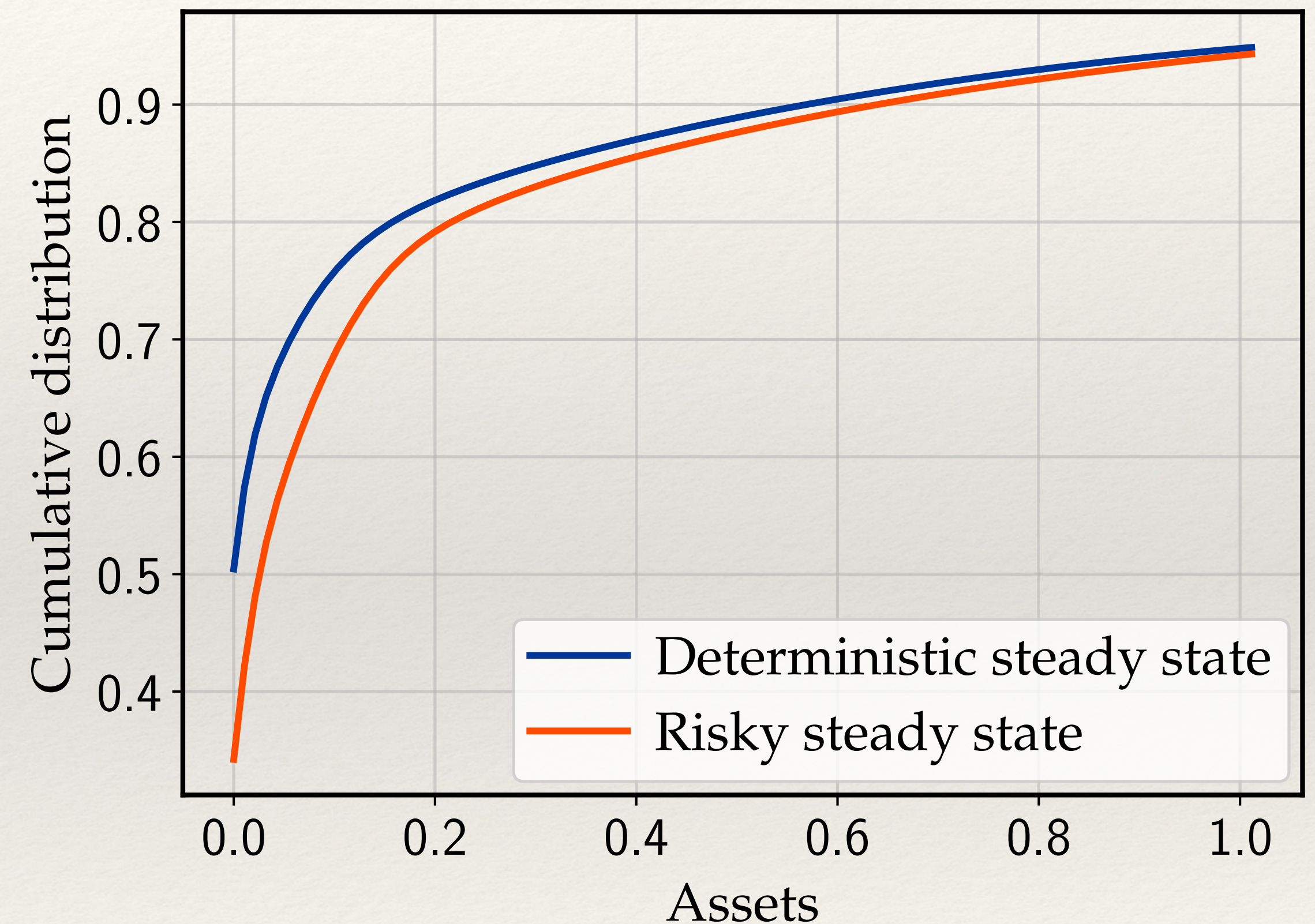
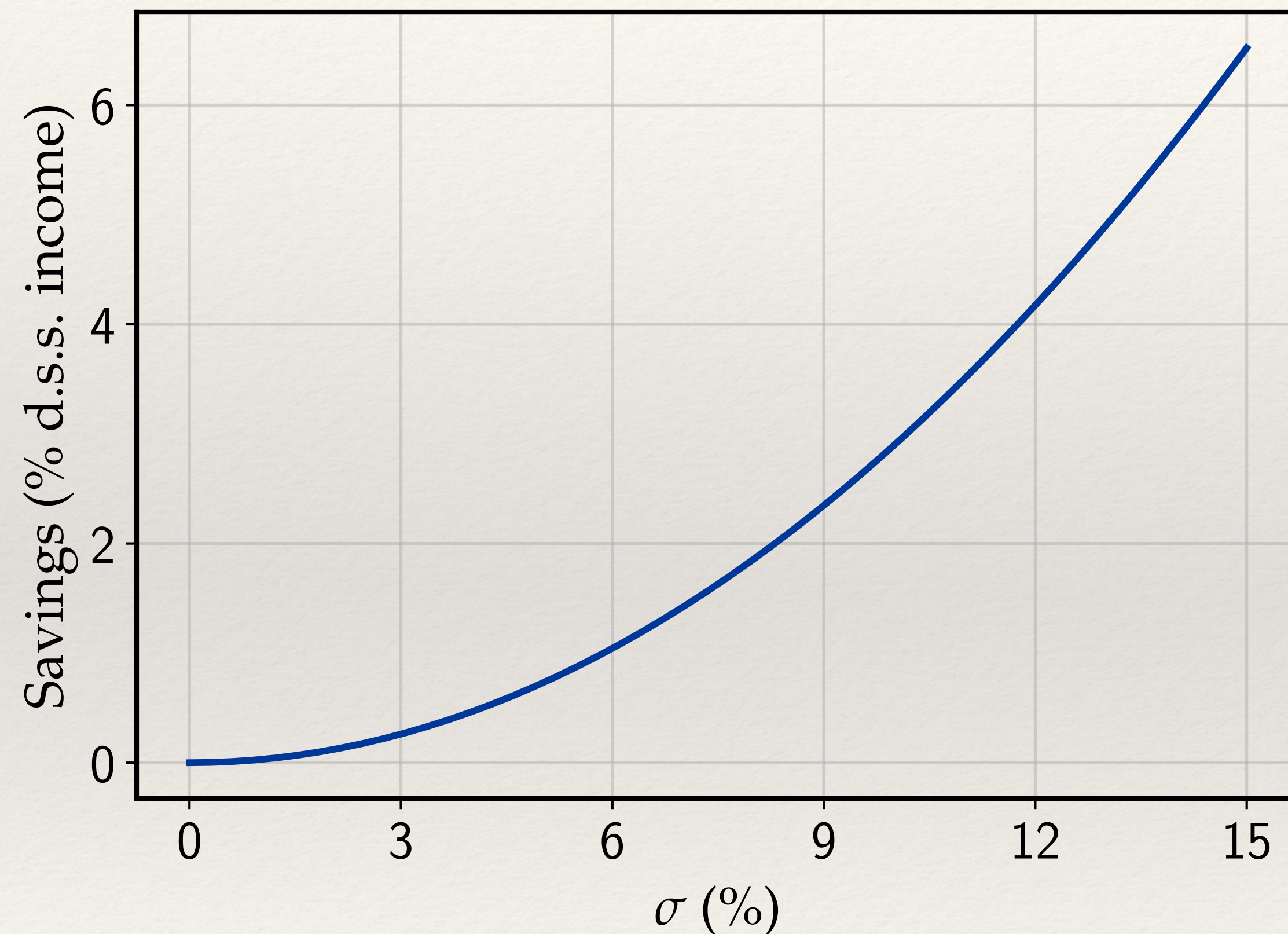
# Application: history dependence in consumption



- ❖ High income shock in past implies more assets today, lower MPCs



# Application: aggregate precautionary savings



- ❖ Perceived aggregate income risk raises risky steady-state savings, lowers MPCs



# Third-Order Certainty Correspondence



# Third-order sequence space perturbation

- ❖ Continue expanding to 3rd order, with symmetry implying  $\mathcal{X}_{\sigma jk} = 0$
- ❖ Obtain **cubic** MA process:

$$X_t \approx X_t^{(2)} + \boxed{\frac{1}{6} \sum_{j=0}^{\infty} \frac{\partial^3 \mathcal{X}}{\partial \epsilon_{-j} \partial \sigma^2} \epsilon_{t-j} \sigma^2} + \boxed{\frac{1}{6} \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} \sum_{l=0}^{\infty} \frac{\partial^3 \mathcal{X}}{\partial \epsilon_{-j} \partial \epsilon_{-k} \partial \epsilon_{-l}} \epsilon_{t-j} \epsilon_{t-k} \epsilon_{t-l}}$$

- ❖ All **triple interactions** can be obtained from 1, 2, or 3 consecutive MIT shocks
- ❖ **New term** includes 2 effects:
  - ❖ How concerns about future risk affects first-order impulse response
  - ❖ How past shocks affect strength of precautionary motive



# New term in $\epsilon_{t-j}\sigma^2$

- ❖ Split the new term into these two effects

$$\frac{\partial^3 \mathcal{X}}{\partial \epsilon_{-j} \partial \sigma^2} = \boxed{\left( \frac{\partial^3 \mathcal{X}}{\partial \epsilon_{-j} \partial \sigma^2} \right)^{rss}} + \boxed{\left( \frac{\partial^3 \mathcal{X}}{\partial \epsilon_{-j} \partial \sigma^2} \right)^{tvps}}$$

Effect of risky steady state  
on impulse response

Effect of shock at date  $-j$   
on precautionary motive at date 0

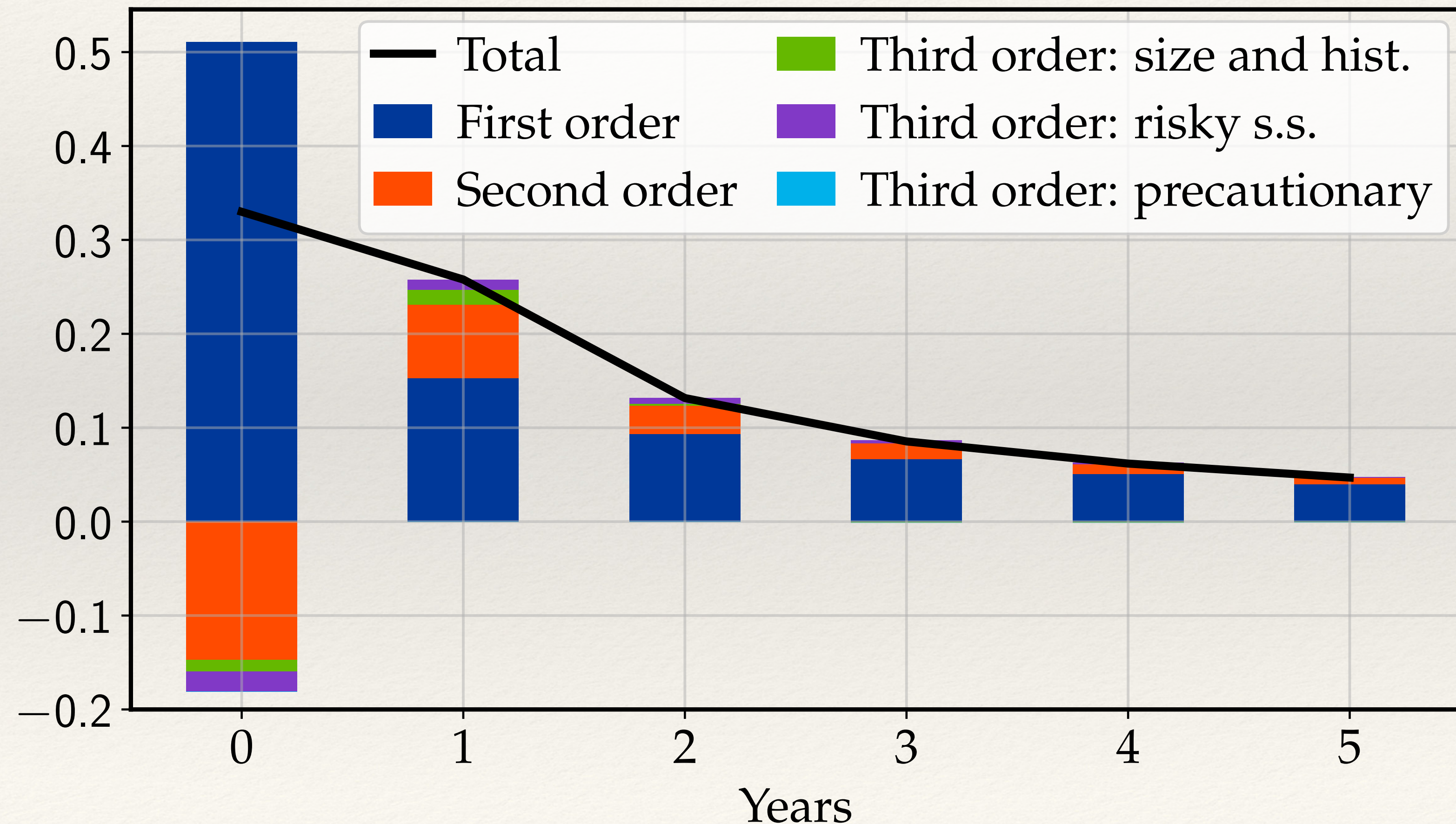
- ❖ Get **1st** using impulse routines starting from the RSS distribution and expectations
- ❖ Get **2nd** by constructing a deterministic sequence  $X_t$  of perturbed expectations:

$$F \left( X_{t-1}, X_t, X_{t+1} + \frac{1}{6} \frac{\partial^3 \mathcal{X}}{\partial \epsilon_{-t} \partial \epsilon_0^2} \sigma^2, 0 \right) = 0$$



# Application: third order iMPCs

❖ Generalized impulse response of  $C_t$  to innovation in income:





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# Summary: certainty correspondence

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- ❖ **Certainty correspondence:** can obtain 2nd and 3rd(+) order perturbation solutions to rational expectation solutions with perfect foresight methods!
  - ❖ ... despite agents being forward looking and anticipating future aggregate risk!
- ❖ Opens up lots of possibilities for research
  - ❖ state-dependent or size-dependent policies (e.g. fiscal or monetary)
  - ❖ welfare cost of business cycles
  - ❖ steady-state and time variation in precautionary savings or risk premia...



# Brief application to GE HANK model



# Speeding up the GE solution using Jacobians

- ❖ Asset market clearing error:  $H_t(\{Y_s\}, \{B_s\}) \equiv A_t(\{Y_s - T_s(B_s)\}) - B_t$
- ❖ Given  $\{B_s\}$ , output solves

$$\mathbf{H}(\mathbf{Y}, \mathbf{B}) = 0$$

- ❖ First-order solution:  $d\mathbf{Y} = -\mathbf{H}_{\mathbf{Y}}^{-1}\mathbf{H}_{\mathbf{B}}d\mathbf{B}$  [ABRS 2021]
- ❖ Size-dependent impulse response, avoiding solving nonlinearly for  $Y_t(\{B_s\})!$

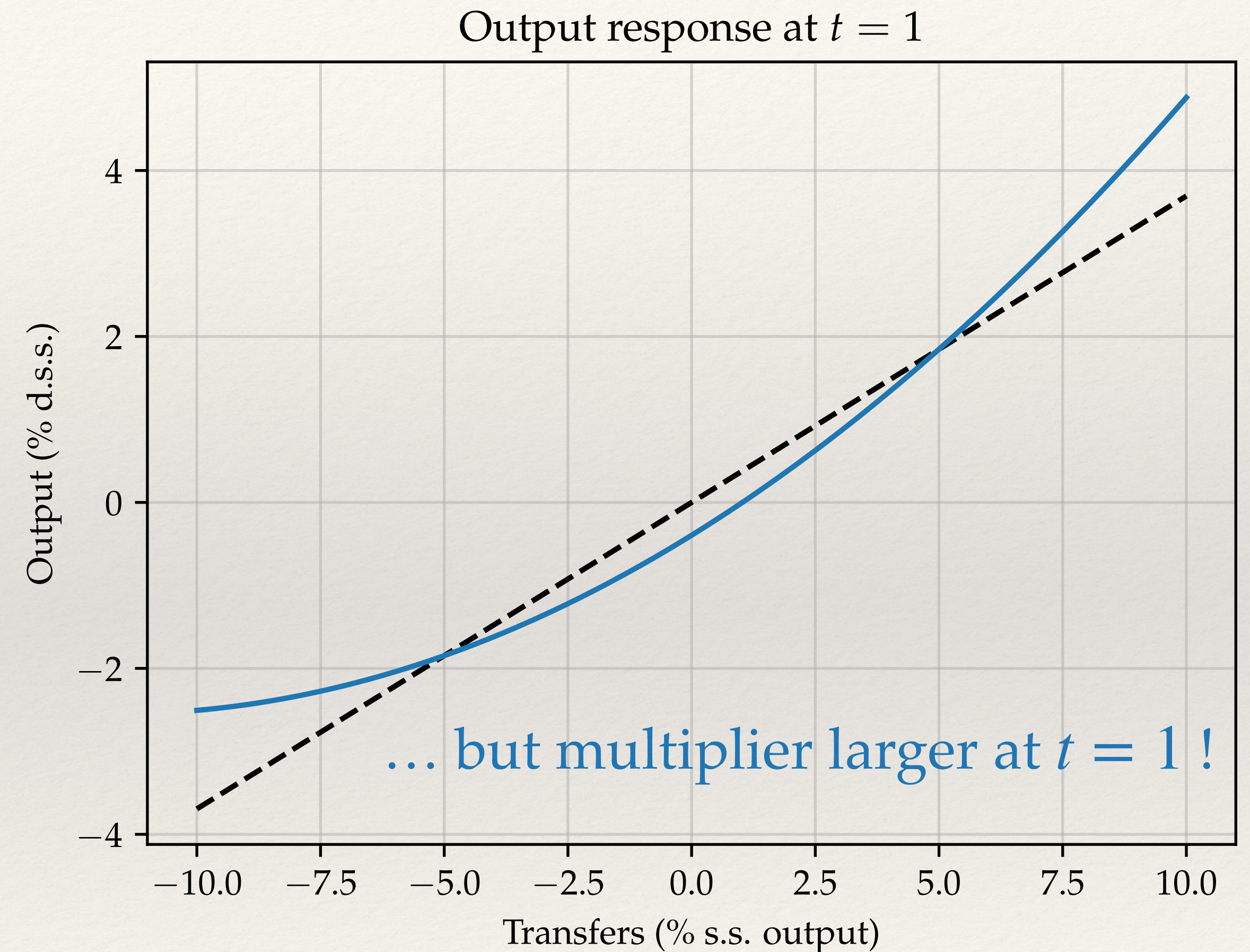
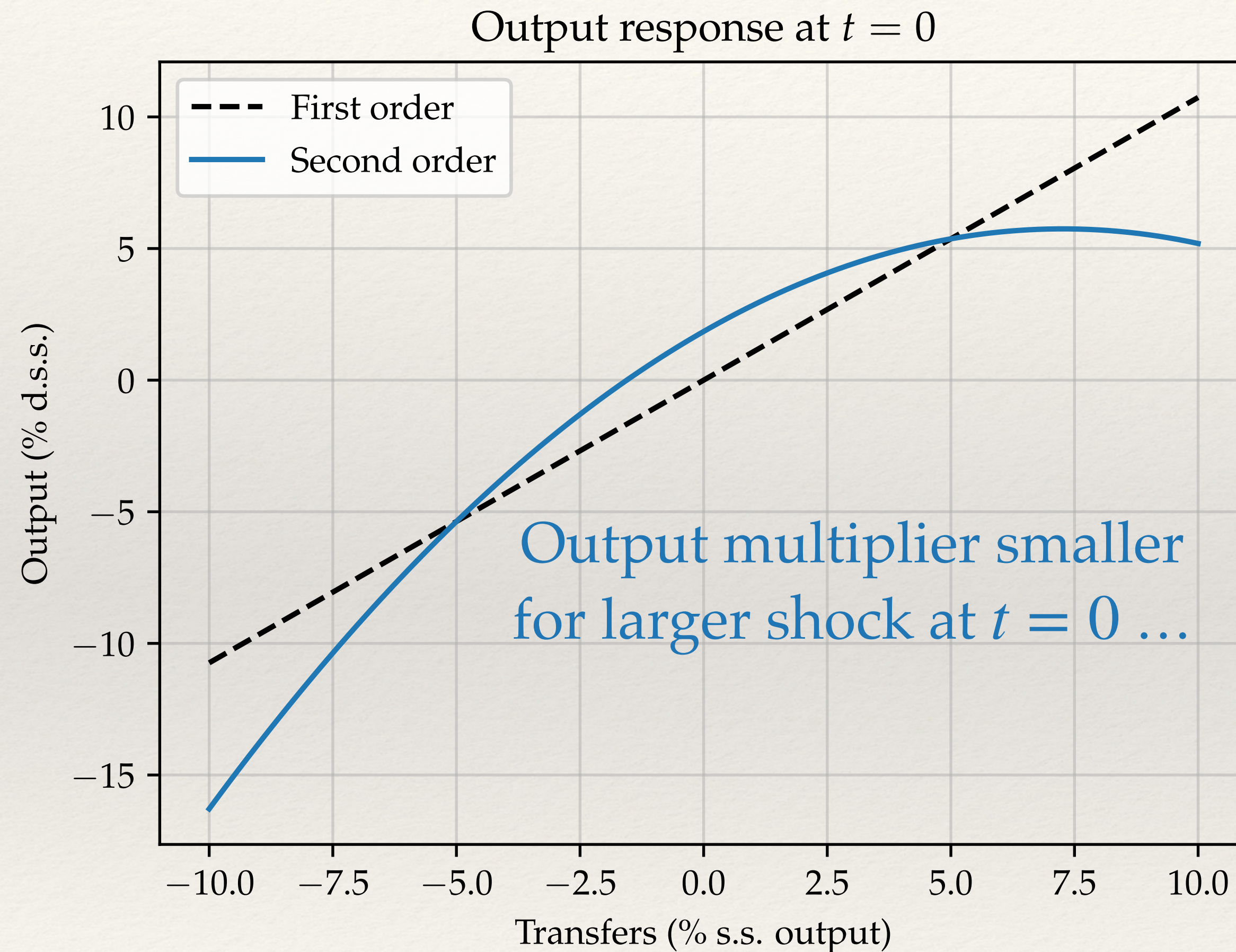
$$d^2\mathbf{Y} = -\mathbf{H}_{\mathbf{Y}}^{-1} \left( \lim_{\nu \rightarrow 0} \frac{\mathbf{H}(Y + \nu d\mathbf{Y}, B + \nu d\mathbf{B})}{\nu^2} \right)$$

same as for  $d\mathbf{Y}$  nonlinear error from  $d\mathbf{Y}$

Use similar approach for state dependence  
(just using different initial distribution)



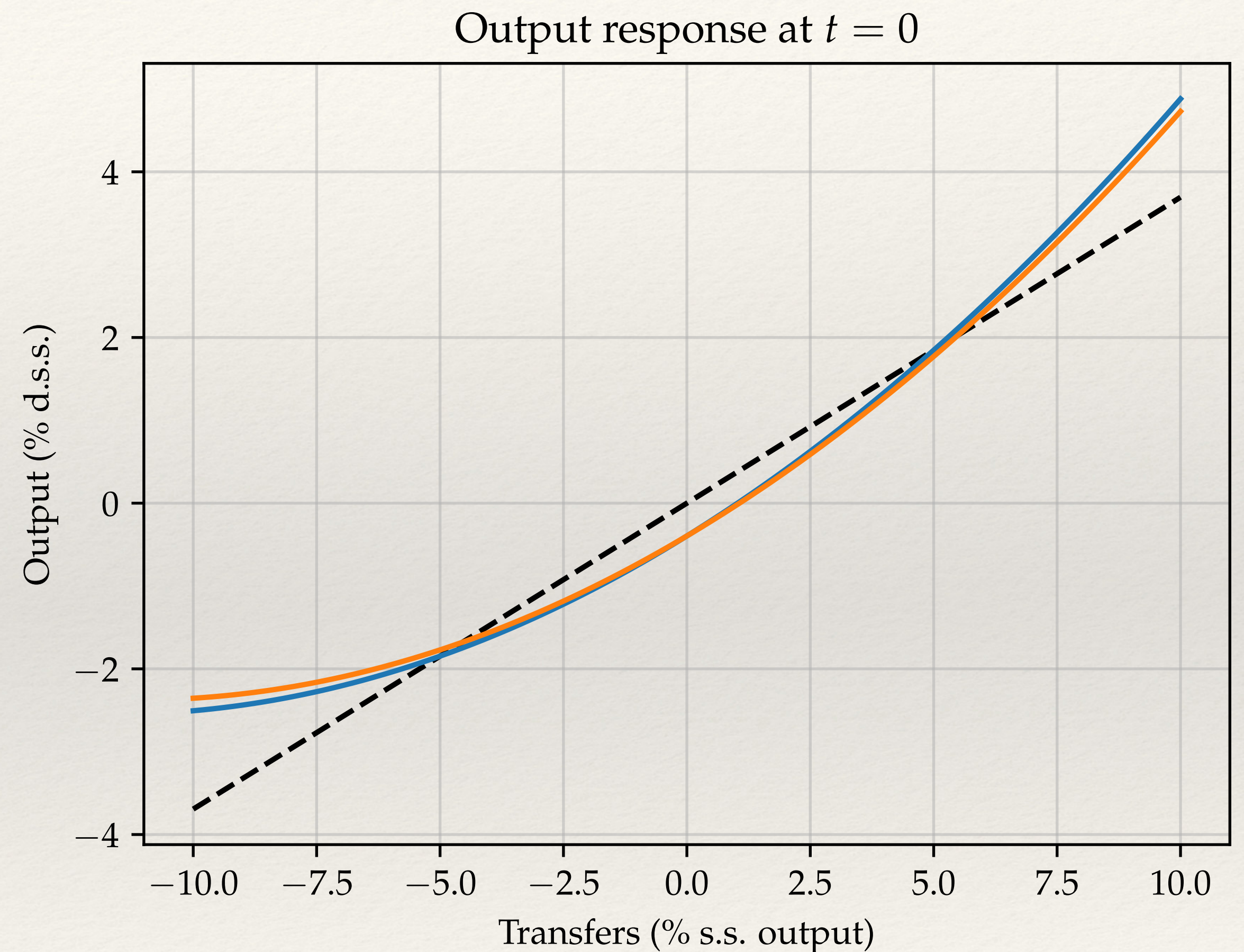
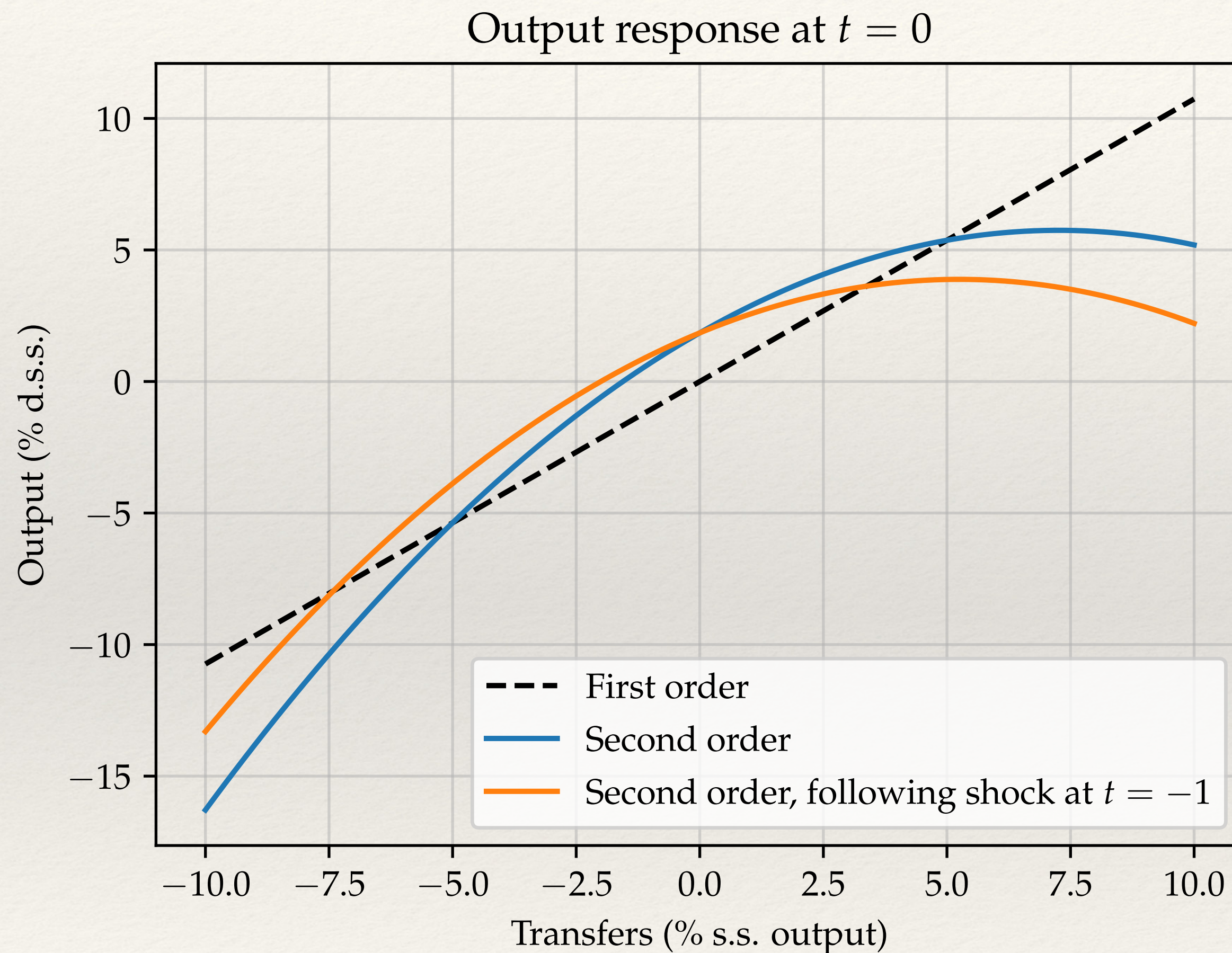
# Generalized impulse response: size dependence



❖ Concavity / convexity of responses translate to GE!



# Generalized impulse response: state dependence



❖ Same with history: past transfers imply more excess savings today, lower MPCs



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# Conclusion

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MIT shocks (on a smooth model) are all you need!